Clear: Goal-Conditioned, Faithful Re-Narration of Web Pages

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Renarration, not Summarization

Renarration reshapes a web page to fit the reader's *goal* and *background knowledge* while preserving its factual meaning. It is not a summary, a translation, or a stylistic rewrite: the output stays faithful, but becomes clearer and genuinely useful for what the reader is trying to do right now.

- Adapts *structure*, *depth* and *complexity* to one explicit reading goal.
- Speaks the reader's language and leans on real-world analogies.
- Reads text *and* images together — figures are explained, not skipped.
- Page-faithful: **nothing invented, nothing silently dropped**.

Why One LLM Call Is Not Enough

Feeding raw page HTML to a single model and asking for a rewrite quietly drops facts on dense pages and can never prove its coverage. Clear splits the work in two:

- **Extract first** — find every fact exhaustively, under a budget.
- **Renarrate second** — turn the *verified* facts into goal-shaped prose.

Separating the phases makes coverage auditable and keeps the rewrite honest.

Phase 1 — Exhaustive Extraction

A planner cuts the page into timeout-safe segments. Parallel *text* subagents do exhaustive fact-finding; *vision* subagents act as curators and captioners, never as fact sources. A hierarchical reducer merges everything (map/reduce) into a single fact ledger.

- Coverage is bounded by a per-run **budget** (wall-clock / tokens / storage), never a fixed count.
- Every skipped item is logged — the UI shows "*N* items skipped" instead of implying full coverage.

Phase 2 — Goal-Conditioned Renarration

The verified facts, the saved reading goal and the active task become natural prose that explains the subject directly. Image captions are re-narrated in parallel into the reader's language. The reading goal itself is produced by a side-panel chat, then steers emphasis, depth and style.

Problem Statement

People constantly meet pages written for *someone else* — a specialist, in another language, at the wrong depth. A summary throws away the parts that matter to them; hours in a chatbot is friction. Clear renarrates the page **in place**: faithful to the source, shaped by the reader's own goal.

How Clear Works — A Two-Phase Pipeline

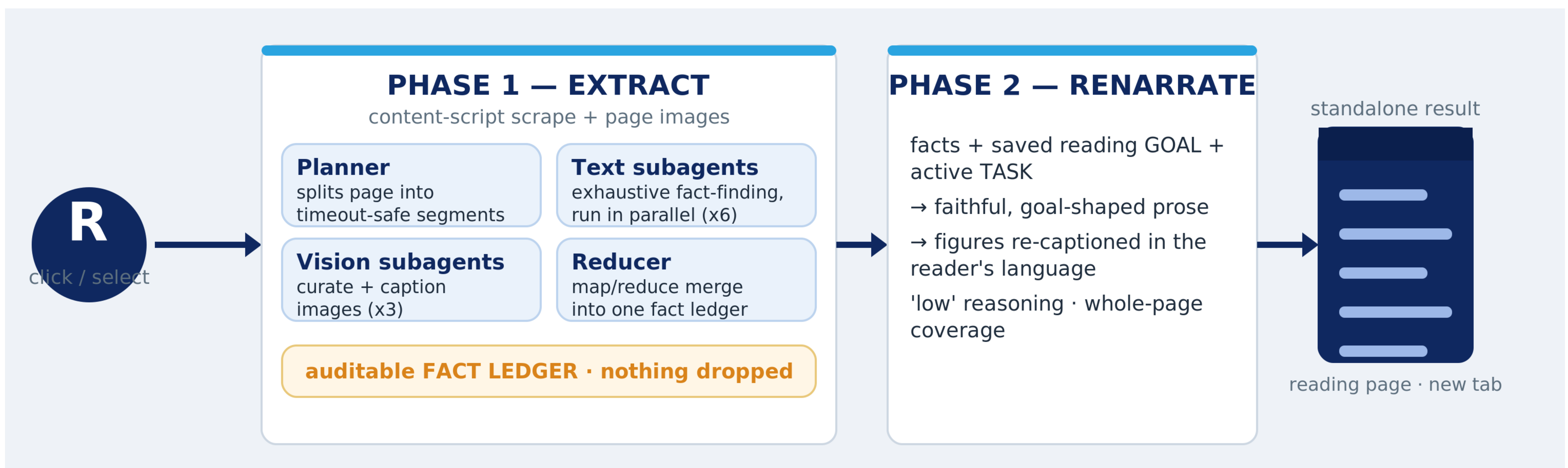
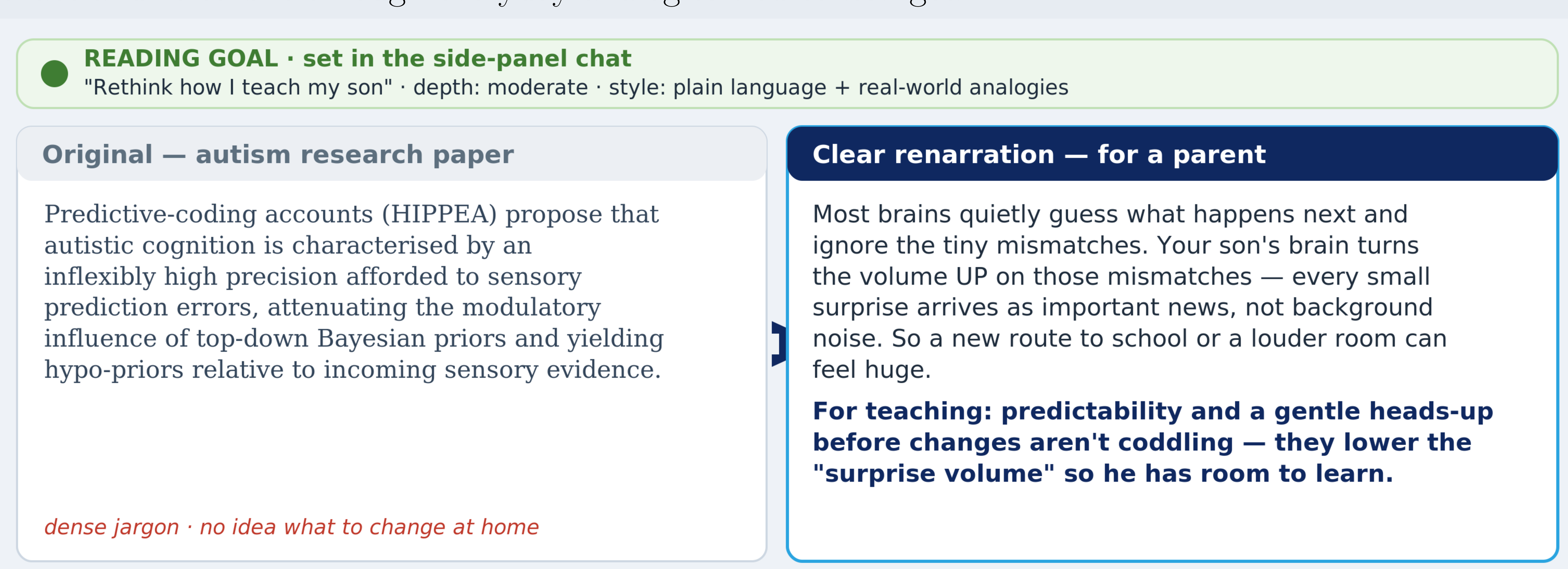


Figure 1:One click: exhaustive extraction into an auditable fact ledger, then goal-conditioned renarration into a standalone reading page.

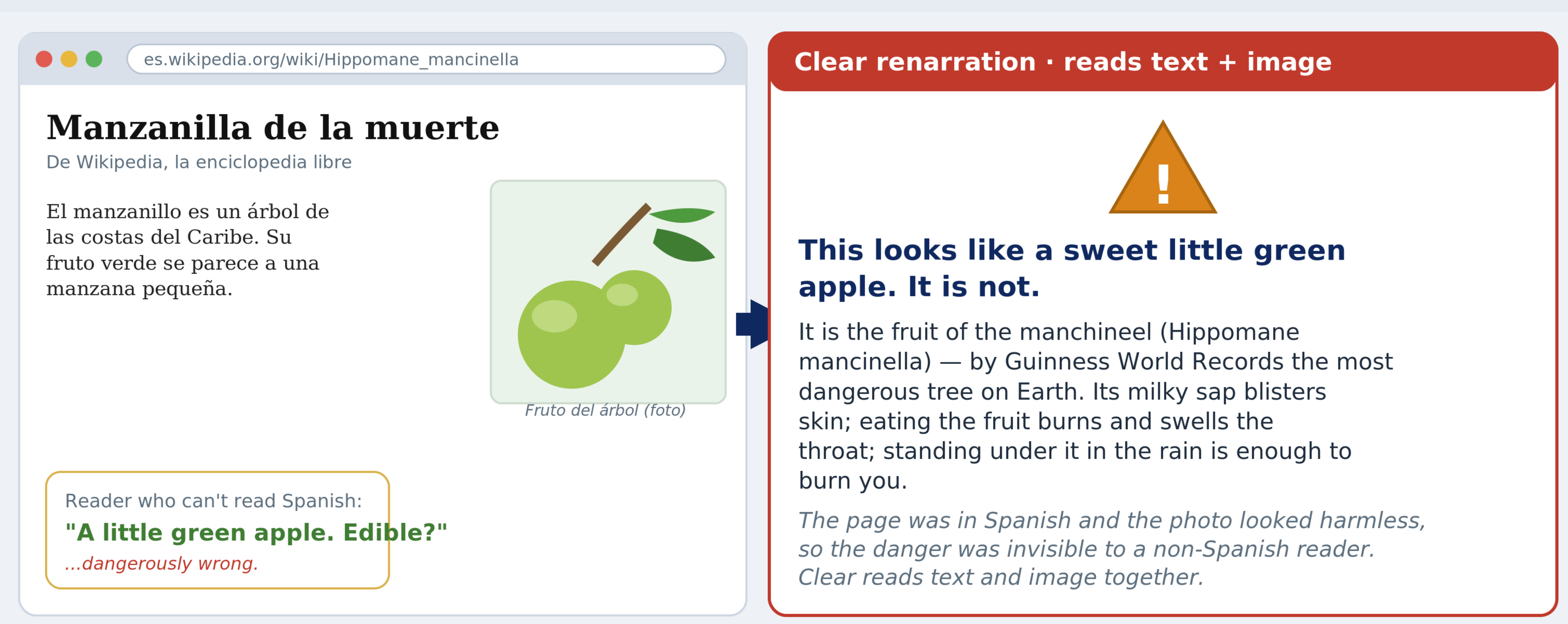
Use Case 1 — A Parent vs. the Research

A mother wants to rethink how she teaches her autistic son using new research, but the papers are a wall of jargon. Clear keeps the science exact and re-tells it through everyday analogies tied to *her* goal.



Use Case 2 — A Photo in Another Language That Lies

A Spanish page shows what looks like a harmless little green apple. A non-Spanish reader would never guess it is the *manchineel* — by Guinness World Records the most dangerous tree on Earth. Clear reads the foreign text and the image **together**.



What Clear Buys You

- **Faithful** — page-grounded; nothing invented, nothing silently dropped.
- **Personal** — one explicit goal reshapes depth, language and analogies.
- **Multimodal** — images are curated and explained, not discarded.
- **Honest about coverage** — budget skips are surfaced, never hidden.

Architecture

- Chrome MV3 extension; background service worker bundled with Vite; no UI framework.
- OpenAI Responses API wrapper — every request uses `store: false` (no server-side retention).
- Budgets keep the worker under the MV3 "unresponsive" kill window.
- The reading page is one self-contained HTML file with images embedded as data URIs — it works offline.

Security Posture

- A build-time API key is extractable from any shipped extension — mitigated by key rotation and budget caps; a server-side proxy / per-user key is the planned fix.
- <all_urls> is required for the on-page overlay and for embedding third-party images.
- The overlay renders inside a **shadow DOM**; final renarrated text is written with `textContent`, never generated HTML.

Future Work

- Move credentials behind a proxy so no usable key ships in the bundle.
- On-device / local models for fully private renarration.
- Reader-controlled analogy depth and persistent per-domain goals.

References

[1] Van de Cruys, S., et al. "Precise minds in uncertain worlds: predictive coding in autism." *Psychological Review* 121.4 (2014): 649–675.
[2] "Hippomane mancinella (manzanilla de la muerte)." Wikipedia. Listed by Guinness World Records as the world's most dangerous tree.
[3] OpenAI. "Responses API." OpenAI Platform Documentation, 2025.
[4] Google. "Chrome Extensions — Manifest V3." Chrome for Developers.

Project: Clear — On-Device Renarration Assistant (Chrome MV3)